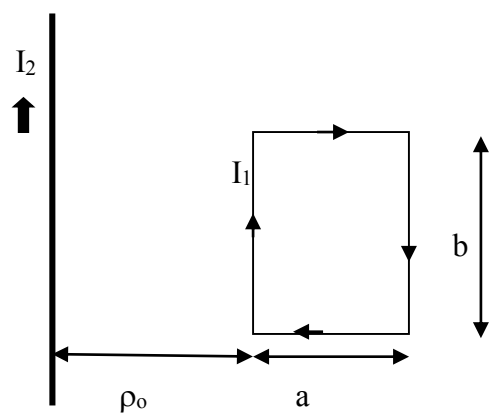
Course PH410 Course coordinator: Alika Khare

Problem Set _08 Due Date: April 10, 2025

1. Griffith third edition, problem nos 5.6, 5.7, 5.9, 5.11, 5.15, 5.16 , 5.36.
2. A very long coaxial cable consists of central wire of radius a carrying a uniformly distributed current I and outer cylindrical shell of inner radius b and thickness t carrying a current $-I$ (return path). Calculate the magnetic field ($\boldsymbol{H}$) in the region $0 \leq \rho \leq a$, $a \leq \rho \leq b$, $b \leq \rho \leq (b+t)$ and $\rho \geq (b+t)$. Plot magnetic field intensity as a function of ρ.
3. A current is flowing in a long conducting wire of radius a. The current is distributed in the wire such that the current density at a distance r from the wire is given by $J = J_o\left(1 + \frac{r^2}{a^2}\right)$. Find the magnetic field inside and outside the wire and the total current through it. Plot the magnetic field as a function of distance from the axis of the wire.
4. A rectangular wire loop of width a and height b is carrying a current I_1 in the clock wise direction is placed at a distance of ρ_o from an infinitely long wire carrying a current I_2 in upward direction as shown. Calculate the force experienced by the loop.



5. A closed wire loop of radius a carries a constant current I. Find the magnetic field any where in space. (Hint: write down the general expression for A in such case, then apply the multipole expansion in terms of Legendre polynomial, as was implemented in electrostatic case and follow the similar steps).

Problem 5.7 For a configuration of charges and currents confined within a volume $\mathcal{V}$, show that

$$\int_{\mathcal{V}} \mathbf{J}\, d\tau = d\mathbf{p}/dt, \tag{5.31}$$

where $\mathbf{p}$ is the total dipole moment. [*Hint:* evaluate $\int_{\mathcal{V}} \nabla \cdot (x\mathbf{J})\, d\tau$.]

$$\vec{p} = \int \rho \vec{r}\, d\tau$$

$$\vec{\nabla}(x\vec{J}) = x(\vec{\nabla}\vec{J}) + \vec{J}\cdot(\vec{\nabla}x)$$

$$\frac{d\vec{p}}{dt} = \frac{d}{dt}\int \rho\vec{r}\, d\tau = \int \frac{\partial \rho}{\partial t}\vec{r}\, d\tau = -\int (\vec{\nabla}\vec{J})\vec{r}\, d\tau$$

$$= -\int (\vec{\nabla}J)(x\hat{x} + y\hat{y} + z\hat{z})\, d\tau$$

$$(\vec{\nabla}\vec{J})x = \vec{\nabla}\cdot(x\vec{J}) - \vec{J}\cdot(\vec{\nabla}x)$$

$$= \nabla\cdot(x\vec{J}) - J_x$$

$$\int (\vec{\nabla}\cdot\vec{J})x\, d\tau = \int \vec{\nabla}\cdot(x\vec{J})\, d\tau - \int J_x\, d\tau$$

$$\frac{d\vec{\rho}}{dt} = -\left[-\int J_x\, d\tau\,\hat{x} - \int J_y\, d\tau\,\hat{y} - \int J_z\, d\tau\,\hat{z}\right] \int_S x\vec{J}\, d\vec{a} - \int_V J_x\, d\tau$$

$$= \int \vec{J}\, d\tau$$

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